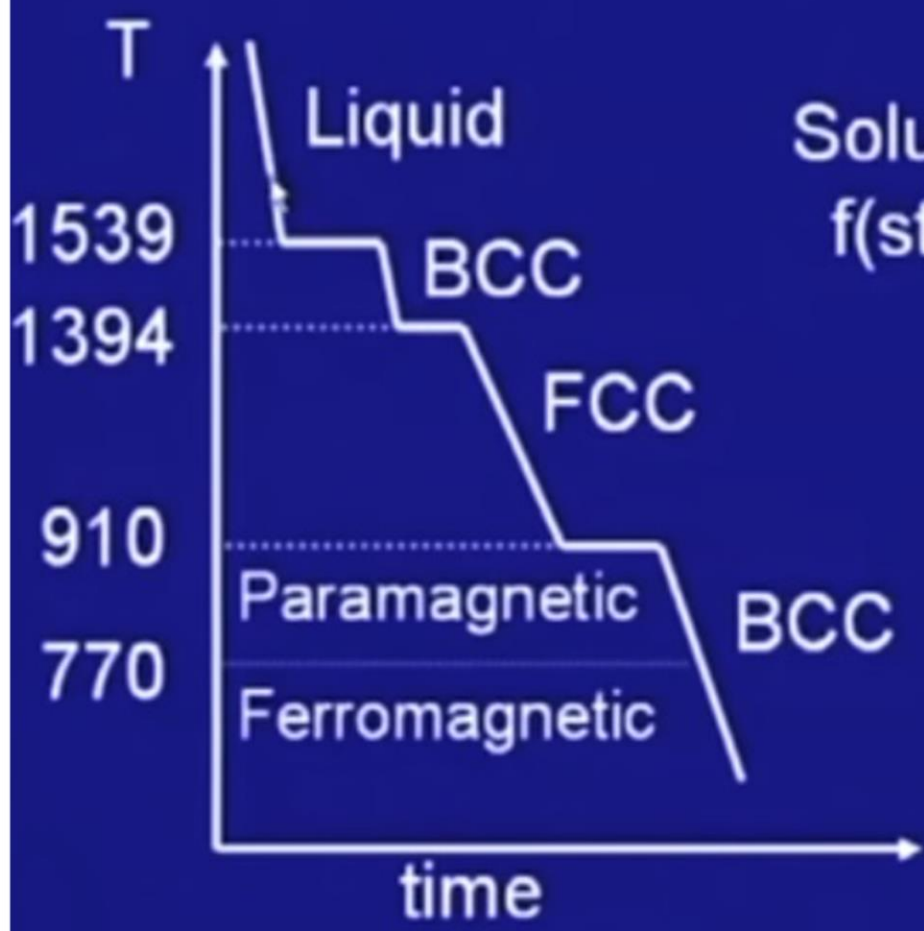
Experiment No:5

Study the effect of Carbon content on the microstructure and mechanical properties of the Steel

Fe : crystal structure



Solubility of carbon in Fe = f(structure, temperature)

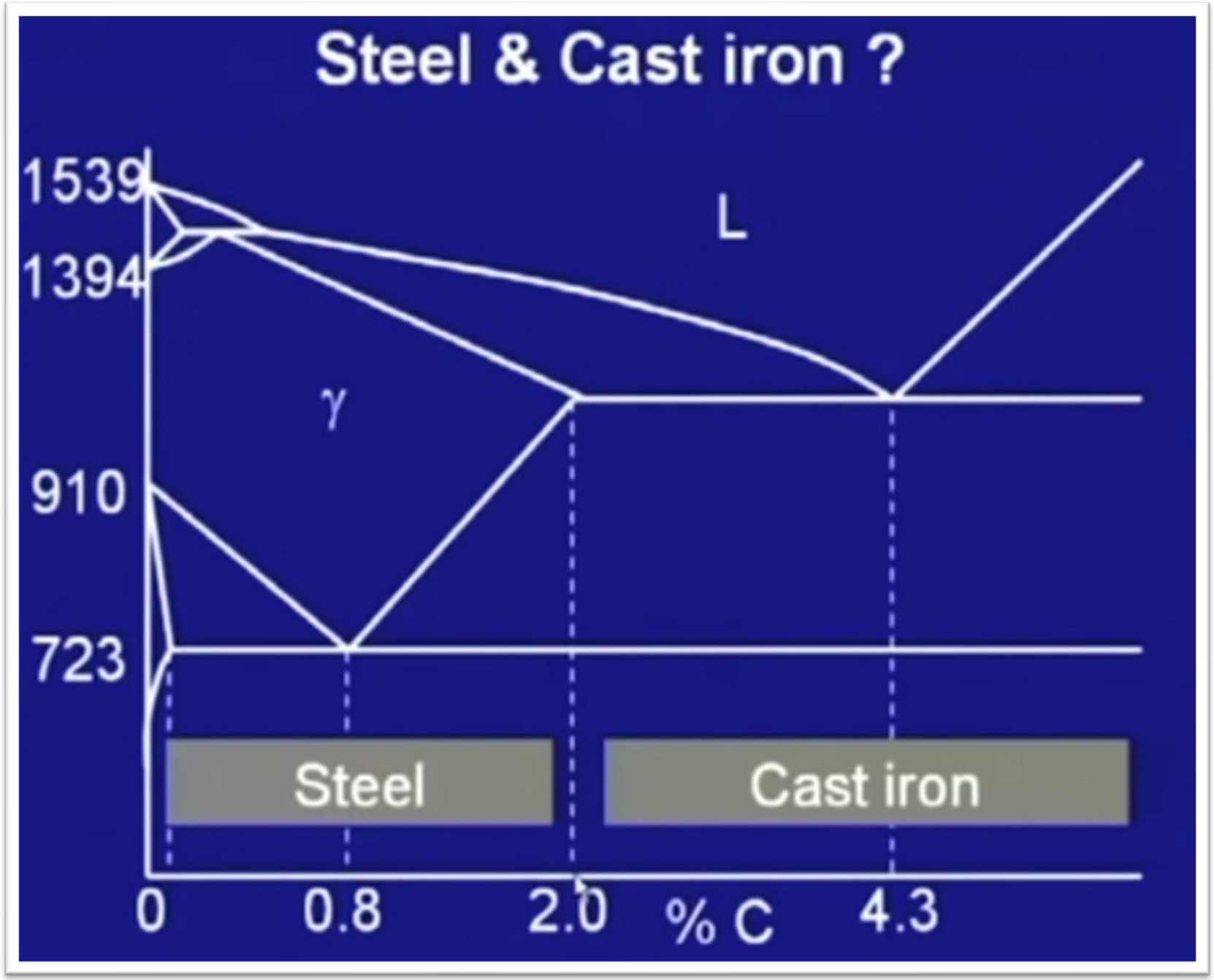
Where is the carbon located in iron lattice?

Invariant reactions in iron – carbon system

δ (0.08% C) + L (0.55% C) = γ (0.18%C) 1495^0 C

L (4.3% C) = γ (2.06%C) + Fe_3C (6.67%C) 1148 C

γ (0.8%C) = α (0.025%C) + Fe_3C (6.67%C) 727 C


Steel & Cast iron ?
1539
1394
910
723
L
γ
Steel
Cast iron
0
0.8
2.0
% C
4.3